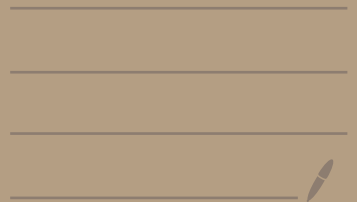


August 4 Gauss Lemma.



We study the irreducible factorizations in $\mathbb{Z}[X]$.

Recall the example of $3x-3 = 3(x-3)$

is an irreducible factorization in $\mathbb{Z}[X]$.

To resolve the pathological example, we introduce

def 1: Let $f(x) = \sum_{i=0}^n a_i x^i \in \mathbb{Z}[x]$

We say $f(x)$ is primitive if the greatest common divisor of all the coefficients is 1.

i.e. the gcd of $a_0, a_1, \dots, a_n = 1$.

eg. $3x-3$ is not primitive

$x^2 - 2x + 1$ is primitive

Lemma 2 : let $f(x) \in \mathbb{Z}[x]$. then.

$\exists ! m \in \mathbb{Z}_+$ s.t $f(x) = p(x)$ where

$p(x)$ is primitive in $\mathbb{Z}[x]$.

proof : Write $f(x) = \sum_{i=0}^n a_i x^i$.

let $m = \gcd(a_0, a_1, \dots, a_n)$

then $f(x) = m \cdot p(x)$ where

$p(x) = \sum_{i=0}^n \frac{a_i}{m} x^i$ is primitive

If $m' \in \mathbb{Z}_+$ and $p'(x)$ primitive

s.t $m' p'(x) = f(x) = m p(x)$

then m' divides $a_i \quad \forall 0 \leq i \leq n$

which implies $m' \mid \gcd(a_0 \dots a_n) = m$.

Suppose $m' \neq m$. Write $p'(x) = \sum_{i=0}^n \frac{a_i}{m'} x^i$.

Then $\gcd\left(\frac{a_0}{m'}, \dots, \frac{a_n}{m'}\right) = \frac{m}{m'} > 1$.

which implies $p'(x)$ not primitive.

So $m' = m$ which implies $p(x) = p'(x)$.

Lemma 3: (Gauss's Lemma)

Let $f(x), g(x) \in \mathbb{Z}[x]$ be primitive

Then $f(x)g(x) \in \mathbb{Z}[x]$ is primitive.

proof: It suffices to show for an

arbitrary prime p . there is some

coefficient of $f(x)g(x)$ such that it is not divisible by p . In particular, there is no prime p that divides all coefficients of $f(x)g(x)$. Therefore, $f(x)g(x)$ is primitive.

Now we prove the claim.

Let p be an arbitrary prime in $\mathbb{Z}$.

$$\text{Write } f(x) = \sum_{i=0}^n a_i x^i$$

$$g(x) = \sum_{j=0}^m b_j x^j$$

Since $f(x), g(x)$ both are primitive,

p does not divide all the coefficients a_i, b_j

Let s be the least index s.t

$p \nmid a_s$ and let t be the

least index s.t $p \nmid b_t$

Then we show p does not divide the

coefficient of x^{s+t} in $f(x)g(x)$.

By construction, that coefficient is

$$\sum_{i+j=s+t} a_i b_j = a_{s+t} b_0 + a_{s+t-1} b_1 + \dots + a_{s+1} b_{t-1} \\ + a_s b_t + a_{s-1} b_{t+1} + \dots + a_0 b_{s+t}$$

Note for $i < s$, $p \mid a_i b_{s+t-i}$

and for $j < t$, $p \mid a_{s+t-j} b_j$.

so p divides each term is $\sum_{i+j=s+t} a_i b_j$

except for $a_s b_t$.

so $p \nmid \sum_{i+j=s+t} a_i b_j$.

Proposition 4: Let $f(x) \in \mathbb{Z}[x]$ be of

positive degree. Suppose $f(x) = g(x)h(x)$

for $g(x), h(x) \in \mathbb{Q}[x]$. then $\exists \bar{g}(x), \bar{h}(x) \in \mathbb{Z}[x]$.

such that $f(x) = \bar{g}(x)\bar{h}(x)$. and $\deg g = \deg \bar{g}$
 $\deg h = \deg \bar{h}$.

proof: If $f(x)$ is not primitive.

by lemma 2, $\exists ! m \in \mathbb{Z}_+$ and primitive

$p(x) \in \mathbb{Z}[x]$ s.t. $f(x) = m \cdot p(x)$.

So it suffices to assume $f(x)$ is primitive.

$$\text{Let } g(x) = \sum_{i=0}^n \frac{p_i}{q_i} x^i = \sum_{i=0}^n a_i x^i \quad p_i, q_i,$$

$$h(x) = \sum_{j=0}^m \frac{p'_j}{q'_j} x^j \quad p'_j, q'_j \in \mathbb{Z}$$

$$= \sum_{j=0}^m b_j x^j$$

$$\text{Then } \exists s, t \in \mathbb{Q} \text{ s.t. } g(x) = s \cdot \bar{g}(x)$$

$$h(x) = t \cdot \bar{h}(x)$$

where $\bar{g}(x), \bar{h}(x)$ primitive in $\mathbb{Z}[x]$.

$$\text{Indeed let } dg = \text{lcm}(q_0, \dots, q_n)$$

$$dh = \text{lcm}(q'_0, \dots, q'_m)$$

$$\text{then } s = dg / \gcd(a_0 dg, \dots, a_n dg)$$

$$t = dh / \gcd(b_0 dh, \dots, b_m dh)$$

Then $f(x) = g(x) h(x) = s.t \bar{g}(x) \bar{h}(x)$.

for $s, t \in \mathbb{Q}$, $\bar{g}(x), \bar{h}(x)$ primitive in $\mathbb{Z}[x]$.

Write $s, t = \frac{u}{v} \in \mathbb{Q}$ for $u, v \in \mathbb{Z}_+$

Then $v \cdot f(x) = u \cdot \bar{g}(x) \bar{h}(x)$ in $\mathbb{Z}[x]$

where $u, v \in \mathbb{Z}_+$, $f(x), \bar{g}(x), \bar{h}(x)$

primitive. Note by Gauss's lemma,

$\bar{g}(x) \bar{h}(x)$ is primitive in $\mathbb{Z}[x]$.

Then by lemma 2. $v = u$

$$f(x) = \bar{g}(x) \bar{h}(x).$$

Corollary 5: Let $f(x) \in \mathbb{Z}[x]$.

Then $f(x)$ is irreducible in $\mathbb{Z}[x]$

iff $f(x)$ is primitive

and $f(x)$ is irreducible in $\mathbb{Q}[x]$.

proof: ($\Rightarrow$) . Assume f irreducible in $\mathbb{Z}[x]$.

Then $f(x)$ primitive. Otherwise, by

Lemma 2: $f(x) = m \cdot p(x)$ is a

nontrivial factorization. By contrapositive

of Proposition 4. f is irreducible in $\mathbb{Q}[x]$

($\Leftarrow$) Suppose f is irreducible in $\mathbb{Q}[x]$.

and f is primitive. As f is irreducible

in $\mathbb{Q}[x]$. $f(x) = m \bar{f}(x)$ for

$m \in \mathbb{Z}_+$. $\bar{f}(x) \in \mathbb{Z}[x]$. As $f(x)$ primitive,

$m=1$. $\bar{f}(x) = f(x)$. So $f(x)$

irreducible in $\mathbb{Z}[x]$.